Protein-Ligand Binding Prediction

Machine Learning Benchmark on the KIBA Dataset

748 proteins · 13,361 ligands · 17,788 interactions

Cold-Both Split · 5 Models · LightGBM ROC-AUC = 0.776

github.com/abuchin/ML_protein_ligand_prediction

Problem Statement

Drug discovery: predicting which small molecules bind to a target protein

▸ Task

Binary classification: does compound X bind to protein Y?
Positive (binder): KIBA score < 12.1 | Negative: KIBA score ≥ 12.1

▸ Dataset

KIBA (Kinase Inhibitor BioActivity) — 1.13M measured & imputed
binding affinities across 4,480 kinases and 683,413 compounds.
This run: 748 proteins · 13,361 ligands · 17,788 interactions (measured only)

▸ Evaluation challenge

Random split → inflated scores (models memorise protein/ligand combos).
Cold-both split → proteins AND ligands unseen at test time → honest estimate
of real-world virtual screening performance.

▸ Key metrics

ROC-AUC (threshold-free ranking) · PR-AUC (imbalance-robust)
EF@1% — Enrichment Factor (drug-discovery standard): how many more
actives in the top 1% than expected by chance?

System Architecture

End-to-end pipeline from raw KIBA scores to trained models

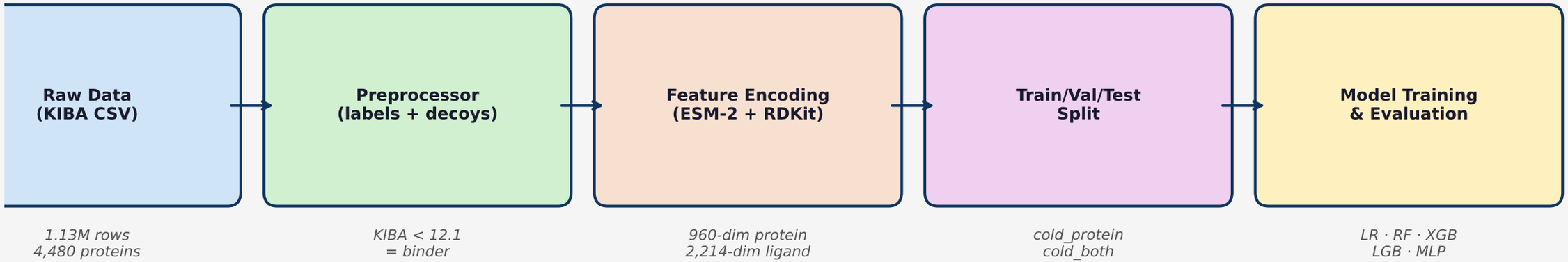
Code structure — src/plbind/

data/preprocessor.py
data/protein_encoder.py (ESM-2)
data/ligand_encoder.py (RDKit)

training/splitter.py
(random / cold_protein /
cold_ligand / scaffold / cold_both)

models/ (LR, RF, XGBoost,
LightGBM, InteractionMLP)
models/base.py (BaseModel API)

evaluation/evaluator.py
evaluation/cold_start.py
(EF, BEDROC, ROC, PR)



Feature Engineering

Protein: ESM-2 language model embeddings · Ligand: Morgan + MACCS + atom-pair + descriptors

Protein features (960 dim)

- ESM-2 esm2_t12_35M_UR50D (Meta, 2022)
- 12-layer transformer, 35M parameters
- mean + max pooling over residues → 2 × 480 = 960 dim
- Pre-trained on 250M UniRef50 sequences
- Cached per UniProt_ID to avoid re-encoding

Auxiliary protein features (~95 dim)

- GO terms (top 50), Pfam domains (top 30)
- Subcellular location, organism one-hot
- Fetched from UniProt REST API

Ligand features (2,214 dim fingerprint + 15 descriptors)

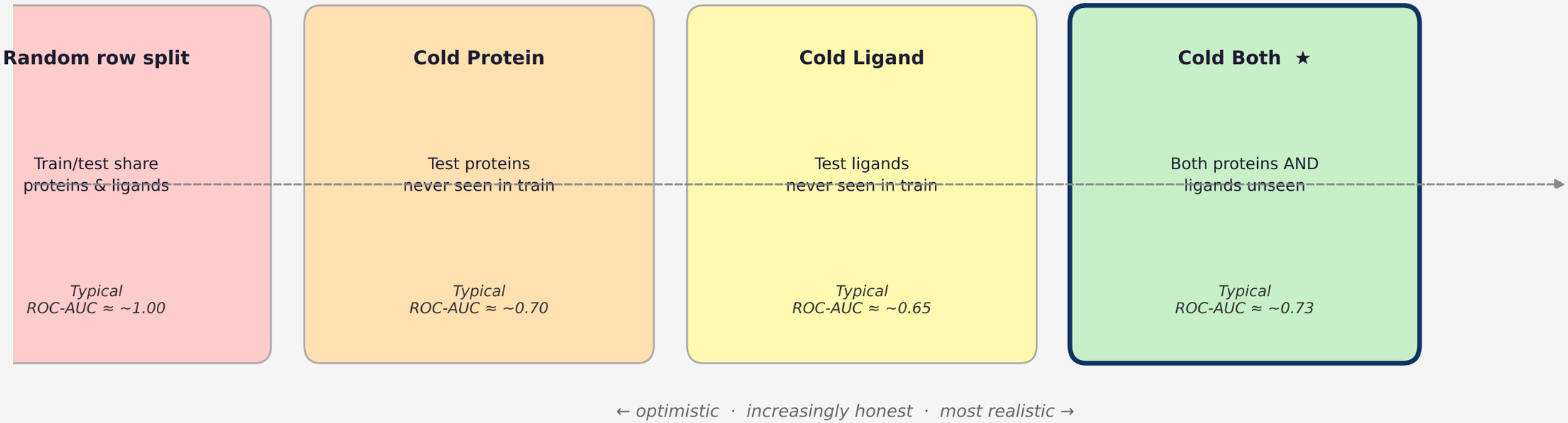
- Morgan count fingerprint radius-2, 1024 bits (ECFP4 counts)
- MACCS keys 166 bits (pharmacophore patterns)
- Atom-pair fingerprint 1024 bits (inter-atom distances)
- 15 physicochemical descriptors: MW, LogP, TPSA, HBD/HBA,
- aromaticity, QED, Gasteiger charges, stereocentres …
- Stored as scipy.sparse (saves ~8 GB vs dense at 500k rows)

InteractionMLP architecture

- Protein → Linear(960, 256) → LayerNorm → ReLU = prot_h
- Ligand → Linear(2229, 256) → LayerNorm → ReLU = lig_h
- interaction_h = prot_h × lig_h (element-wise product)
- concat([prot_h, lig_h, interaction_h]) → 768 → 512 → 256 → 2
- FocalLoss (γ=2, α=0.25), Adam, ReduceLROnPlateau, MPS/CUDA

Split Strategy

cold_both is the honest drug-discovery benchmark — proteins AND ligands unseen at test time



This run used cold_both: 748 proteins · 13,361 ligands · 17,788 interactions → 2,646 test rows (both protein and ligand IDs unseen during training)

Model Results — Cold-Both Split

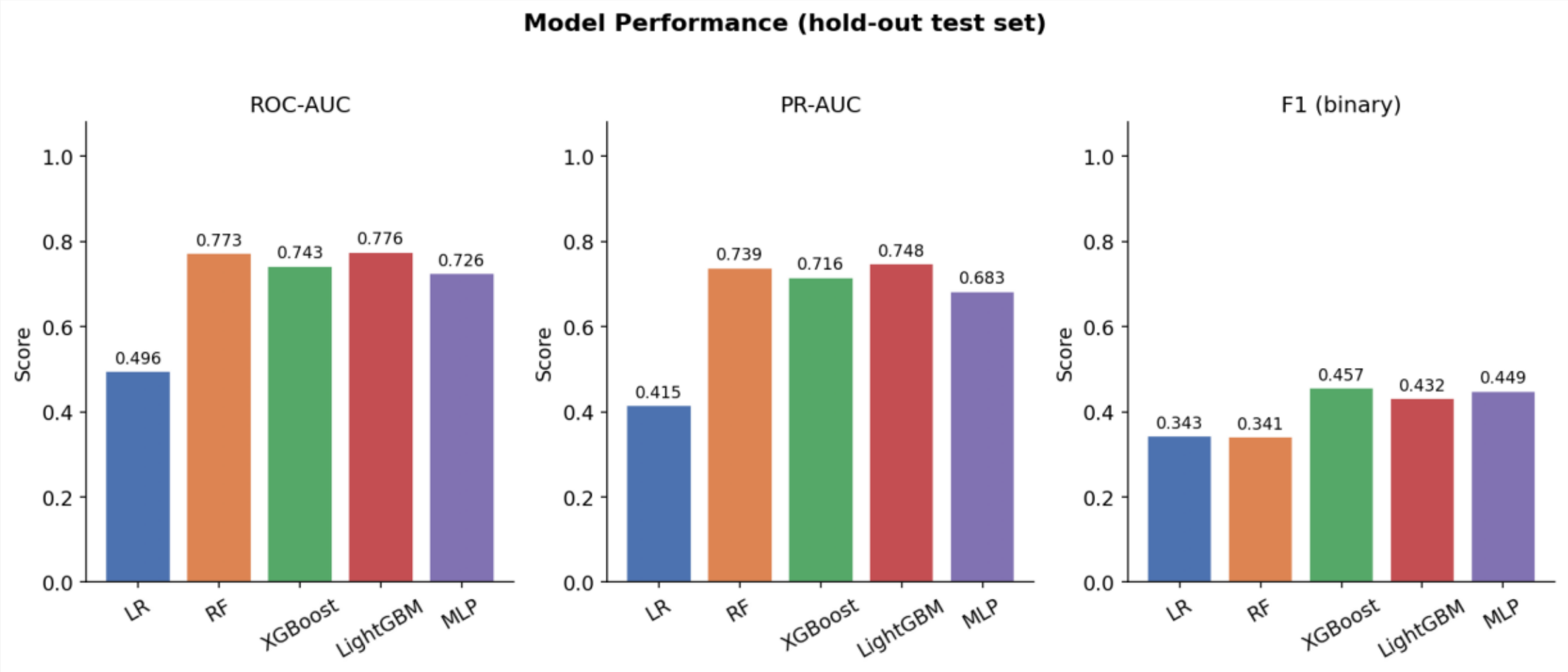
Both proteins and ligands unseen at test time · 748 proteins · 2,646 test rows

Model	ROC-AUC	PR-AUC	F1	Accuracy	EF@1%	EF@5%
1. LightGBM	**0.776**	**0.748**	0.432	**0.684**	**2.375**	**2.340**
2. Random Forest	0.773	0.739	0.341	0.660	2.199	2.215
3. XGBoost	0.743	0.716	**0.457**	0.684	2.287	2.232
4. InteractionMLP	0.726	0.683	0.450	0.677	2.199	2.143
5. Logistic Regression	0.496	0.415	0.343	0.531	0.880	0.929

Cross-validation ROC-AUC (LR, protein-aware StratifiedGroupKFold, 5-fold): 0.467 ± 0.020

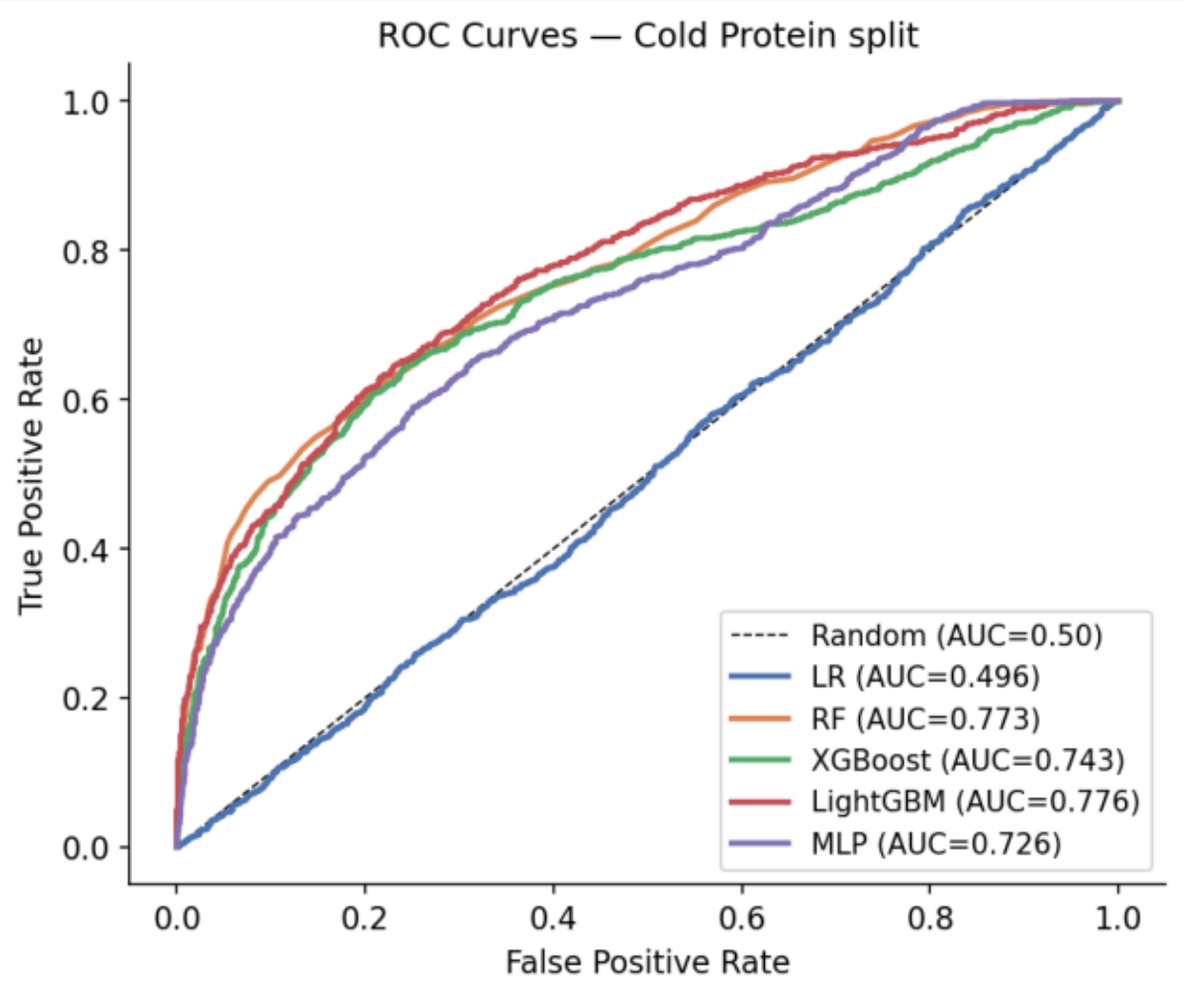
Metric Comparison — Hold-out Test Set

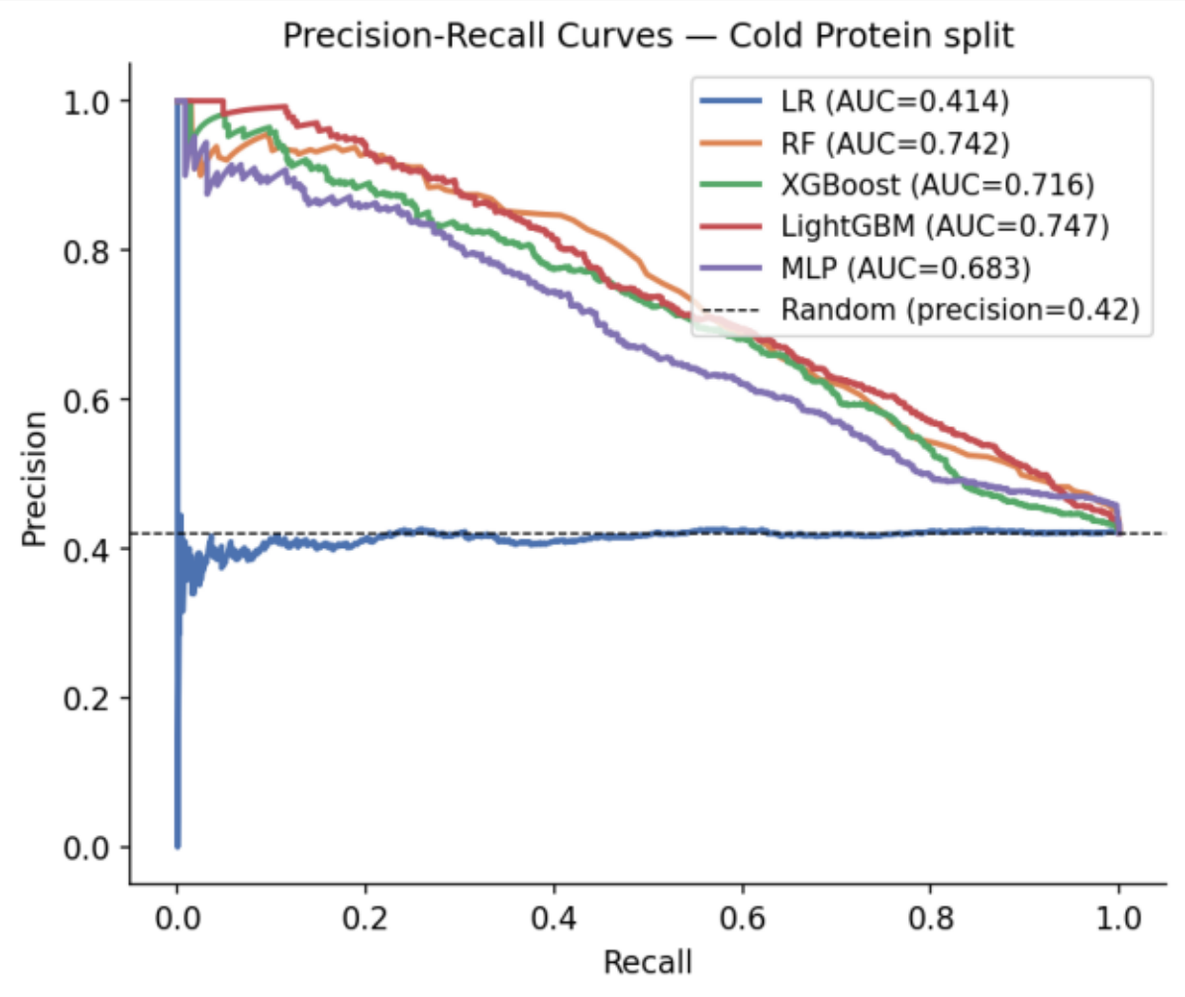
ROC-AUC / PR-AUC / F1 across all models on the cold-both test partition



ROC & Precision-Recall Curves

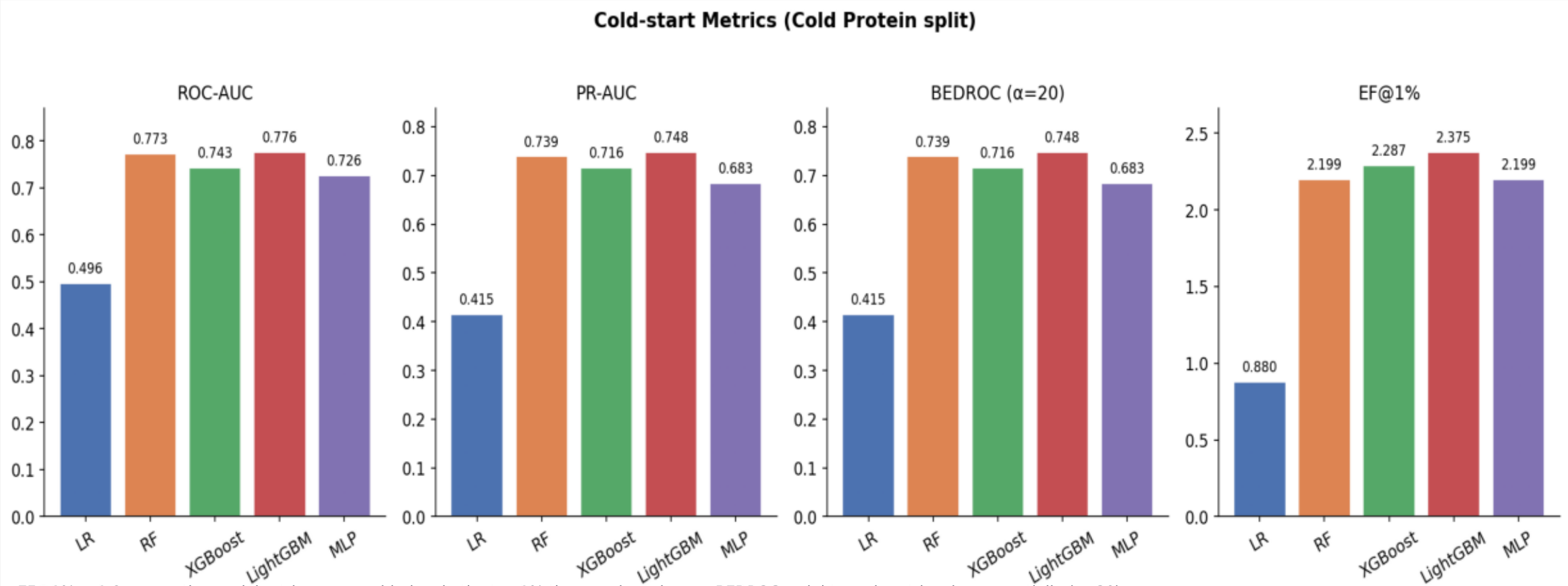
Cold-both split — 2,646 test rows — completely unseen proteins and ligands





Drug-Discovery Metrics

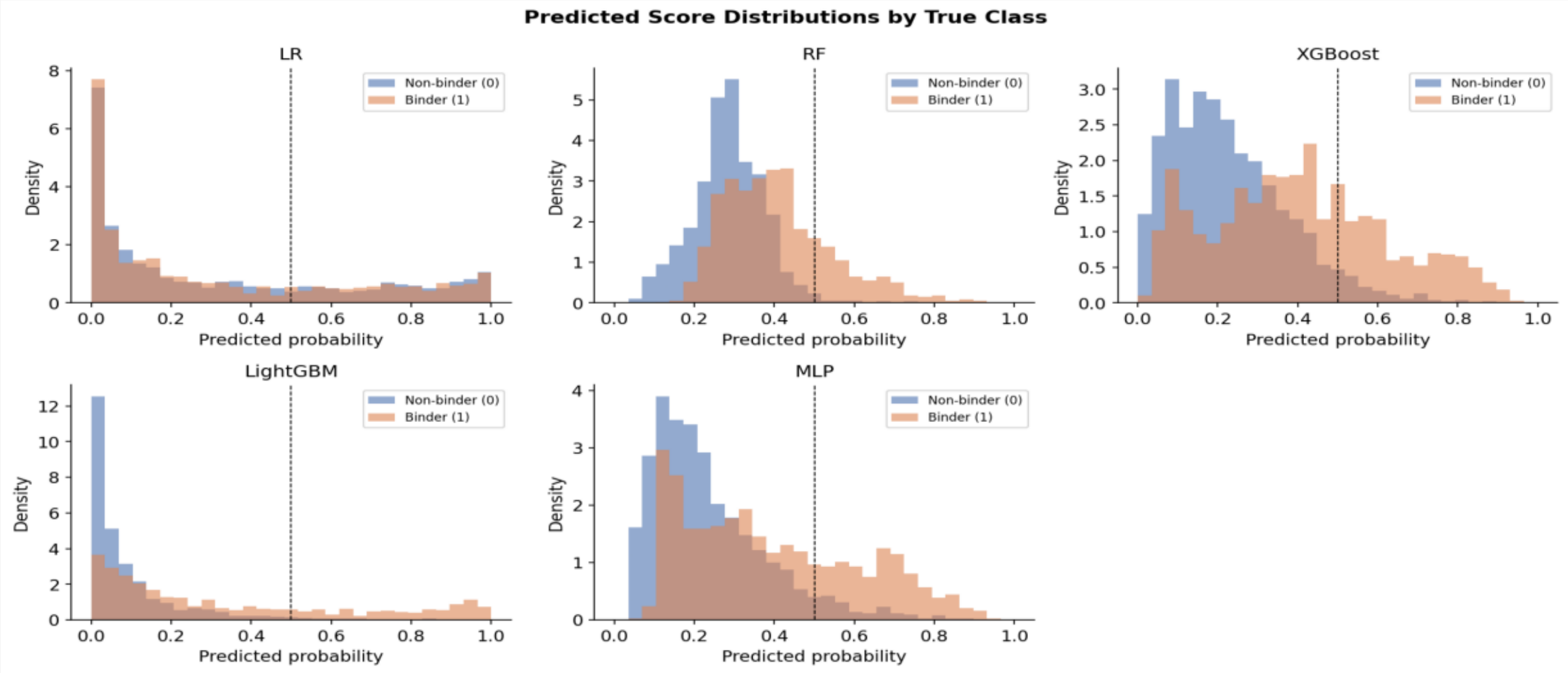
ROC-AUC · PR-AUC · BEDROC · EF@1% — cold-both partition



EF@1% > 1.0 means the model retrieves more binders in the top 1% than random chance. BEDROC weights early retrieval exponentially (α=20).

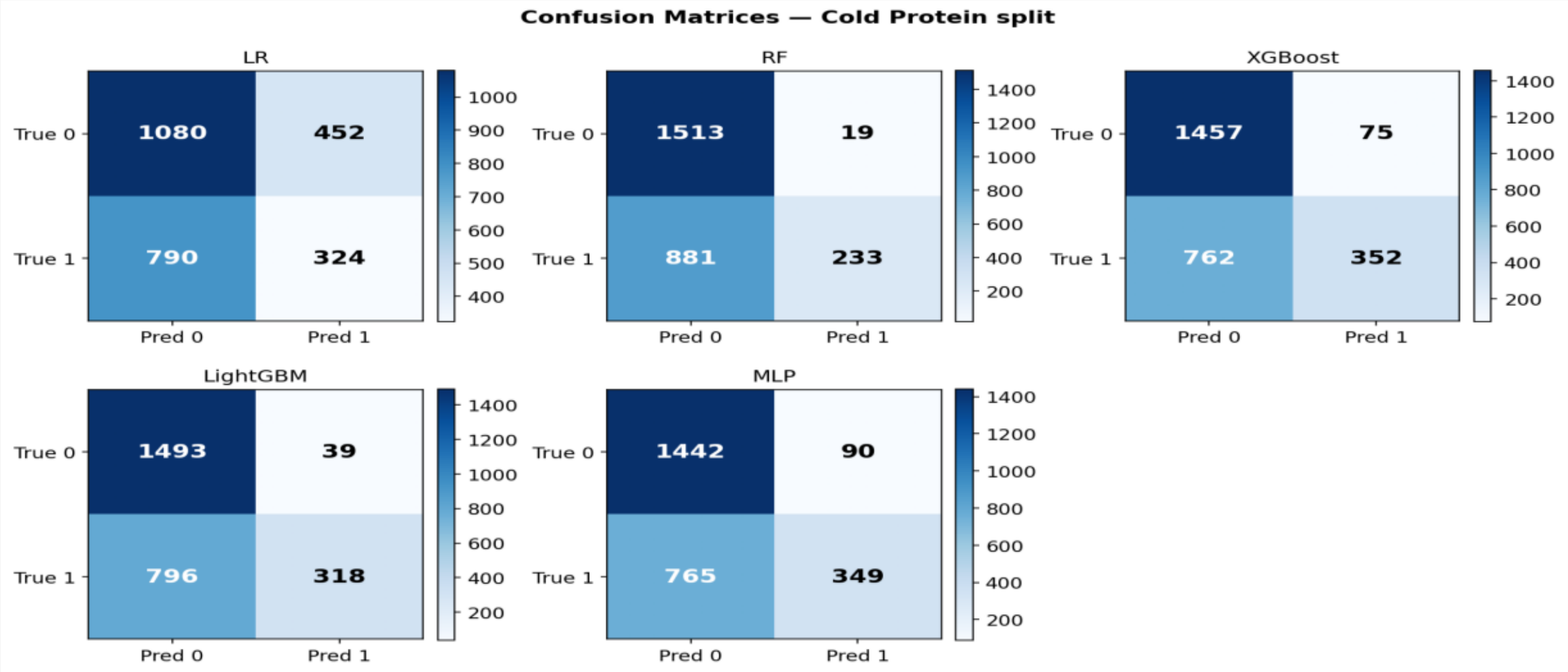
Predicted Score Distributions

Histogram of model output probabilities split by true class (binder vs non-binder)



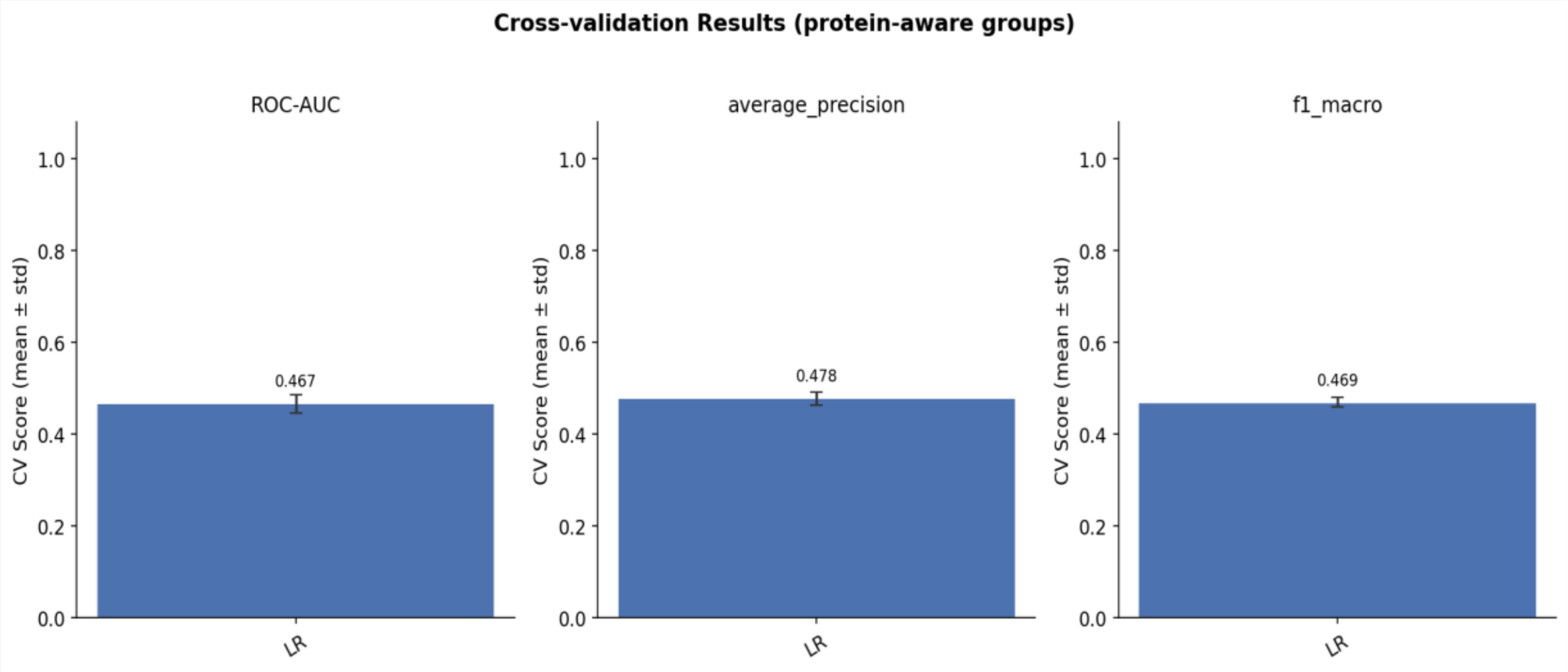
Confusion Matrices

Threshold = 0.5 · cold-both test set



Cross-Validation Results

Protein-aware StratifiedGroupKFold (5-fold) — no protein leakage between folds



Key Takeaways

Cold-Both Benchmark · 748 Proteins · 13,361 Ligands

[1] LightGBM wins cold-both

ROC-AUC = 0.776 | PR-AUC = 0.748 | EF@1% = 2.38x
Tree models outperform LR and MLP on this dataset -- interaction features are non-linear but the tabular fingerprint representation favours ensemble trees.

[2] Enrichment Factor confirms screening utility

EF@1% of 2.2-2.4x means the top-ranked 1% of compounds contains 2x more binders than expected by chance -- practically useful for virtual screening.

[3] Logistic Regression fails on cold-both

ROC-AUC ~ 0.50 -- essentially random for completely unseen pairs.
The binding signal is fundamentally non-linear; linear models cannot generalise.

[4] InteractionMLP is competitive but not dominant

ROC-AUC = 0.726 with only 50 epochs on a 35M-parameter ESM-2 backbone.
A larger ESM-2 (650M, t33) or more training data would likely close the gap with LightGBM.

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[5] Cold-both is the honest benchmark

Random-split ROC-AUC ~ 1.0 -> cold-both ~ 0.77: the 23-point drop is the realistic estimate of how much harder truly novel protein-ligand pairs are.